Federico Bobbio

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RESEARCH INTERESTS

Algorithmic mechanism design and market design; matching theory; mixed-integer and nonlinear optimization; computational complexity; spectrum markets and satellite spectrum policy.

ACADEMIC APPOINTMENTS

2026–present **Assistant Professor**, SKEMA Business School, Lille, France
Globalisation Academy; Centre for Analytics and Operations

2024–2026 **Postdoctoral Researcher**, Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL, USA
Advisors: Randall Berry and Michael Honig (Northwestern University), Rakesh Vohra (University of Pennsylvania), Thanh Nguyen (Purdue University), Vijay Subramanian (University of Michigan)
Affiliations: Communications and Networking Lab, Northwestern University; SpectrumX, an NSF Spectrum Innovation Center

2024 **Research Consultant**, CERC Data Science for Real-Time Decision-Making, Polytechnique Montréal, Canada
Supervisor: Andrea Lodi | *Affiliation:* CIRRELT

2017–2018 **Research Assistant**, Department of Decision Sciences, Bocconi University, Milan, Italy
Supervisor: Pierpaolo Battigalli | *Project:* Reasoning About Strategic Interaction and Emotions

EDUCATION

2024 **Ph.D.**, Department of Computer Science and Operations Research, Université de Montréal, Canada
Thesis: Dynamic Capacities and Priorities in Stable Matching (excellence mention)
Advisor: Margarida Carvalho
Affiliations: CERC Data Science for Real-Time Decision-Making; FRQ–IVADO Research Chair in Data Science for Combinatorial Game Theory; CIRRELT

2017 **M.Sc. in Theoretical Mathematics (Logic)**, *summa cum laude*, University of Pisa, Italy
Thesis: Rationality of Regret in Strategic Games | *Advisor:* Alessandro Berarducci

2016 **Visiting student** (fall term), Institute for Logic, Language and Computation, University of Amsterdam, Netherlands
Advisor: Johan van Benthem

PUBLICATIONS

Journal articles

2026 **F. Bobbio**, M. Carvalho, A. Lodi, I. Rios, A. Torrico. Capacity Planning in Stable Matching. *Operations Research* 74(3):1562–1580.
Extended abstract in the Proceedings of the 24th ACM Conference on Economics and Computation (EC'23), p. 295 (26% acceptance rate).
Runner-up, Open Student Paper Competition, Canadian Operational Research Society, 2023
doi:10.1287/opre.2023.0386 | EC'23: doi:10.1145/3580507.3597771 | arXiv:2110.00734

2026 **F. Bobbio**, M. Carvalho, A. Lodi, A. Torrico. Capacity Variation in the Many-to-one Stable Matching. *Operations Research Letters* 66:107416.
doi:10.1016/j.orl.2026.107416 | arXiv:2205.01302

2023 Y. Zhang, L. Bliek, P. da Costa, R. Refaei Afshar, R. Reijnen, T. Catshoek, D. Vos, S. Verwer, F. Schmitt-Ulms, A. Hottung, T. Shah, M. Sellmann, K. Tierney, C. Perreault-Lafleur, C. Leboeuf, **F. Bobbio**, J. Pepin, W. Almeida Silva, R. Gama, H. L. Fernandes, M. Zaefferer, M. López-Ibáñez, E. Irurozki. The First AI4TSP Competition: Learning to Solve Stochastic Routing Problems. *Artificial Intelligence* 319:103918.
First place (three-way tie), Track 1, AI for TSP Competition at IJCAI 2021 (team prize)
doi:10.1016/j.artint.2023.103918 | arXiv:2201.10453

Refereed conference proceedings

2025 **F. Bobbio**, R. Berry, M. Honig, T. Nguyen, V. Subramanian, R. Vohra. Costly Measurements to Incentivize Spectrum Sharing. *2025 IEEE International Symposium on Dynamic Spectrum Access Networks (DySPAN)*, pp. 1–8.
Best Paper Award, Policy Track
doi:10.1109/DySPAN64764.2025.11115961

2025 I. Rios, **F. Bobbio**, M. Carvalho, A. Torrico. Stable Matching with Contingent Priorities. *Proceedings of the 26th ACM Conference on Economics and Computation (EC'25)*, p. 576 (extended abstract).
Poster at ACM EAAMO'23.
doi:10.1145/3736252.3742596 | arXiv:2409.04914

2018 **F. Bobbio**, J. Cui. A Plausibility Model for Regret Games. In *Multi-Agent Systems and Agreement Technologies (EUMAS 2017)*, Lecture Notes in Artificial Intelligence 10767, Springer, pp. 187–200.
doi:10.1007/978-3-030-01713-2_14

Working papers

2026 **F. Bobbio**, R. Berry, M. Honig. Market Signals without Mission Distortion: Auction Design for Scientific Spectrum Incumbents. Presented at TPRC54, the Research Conference on Communications, Information and Internet Policy.

2026 R. Berry, **F. Bobbio**, M. Ghosh, D. Guo, M. Honig, P. Krishnamurthy, I. Murtazashvili. Overlapping Rights in Orbit: Modeling Aggregate Interference and Spectrum Governance for Satellite Megaconstellations. Presented at TPRC54.

2025 **F. Bobbio**, R. Berry, M. Honig, T. Nguyen, V. Subramanian, R. Vohra. Sharing with Frictions: Limited Transfers and Costly Inspections. arXiv:2512.21793.

2024 **F. Bobbio**, M. Carvalho. Balancing Fairness and Match Quality. Poster at ACM EAAMO'24.

in progress **F. Bobbio**, R. Berry, M. Honig, T. Nguyen, V. Subramanian, R. Vohra. Property Rights and LEO Satellite Competition.

Other publications

2022 W. Almeida Silva, **F. Bobbio**, F. Caye, D. Liu, J. Pepin, C. Perreault-Lafleur, W. St-Arnaud. Design and Implementation of an Heuristic-Enhanced Branch-and-Bound Solver for MILP. arXiv:2206.01857.
Report on the method submitted to the MIP 2022 Computational Competition; poster at the Mixed Integer Programming Workshop 2022.
Outstanding Student Submission (honorable mention), MIP 2022 Computational Competition

2020 **F. Bobbio**, M. Carvalho, A. Nabli, S. Sankaranarayanan, S. Germain, M. O. Hassan, B. Ndao, G. Pagé, J. Piche-Bisson, V. Purenne, R. Shaul. Optimizing the Design of a Loyalty Program. In *Proceedings of the Ninth Montréal Industrial Problem Solving Workshop* (August 2019), Les Cahiers du GERAD G-2020-57.

GRANTS, AWARDS, AND HONORS

Awards

2025 **Best Paper Award**, Policy Track, IEEE DySPAN 2025
for *Costly Measurements to Incentivize Spectrum Sharing*

2023 **Runner-up**, Open Student Paper Competition, Canadian Operational Research Society (CORS)
for *Capacity Planning in Stable Matching*

2022 **Outstanding Student Submission** (honorable mention), MIP 2022 Computational Competition; team leader

2021 **First place** (three-way tie), Track 1, AI for TSP Competition at IJCAI 2021 (team prize)

Scholarships and fellowships

2023 Excellence Scholarship for Thesis Writing, CIRRELT

2020, 2021 Excellence Scholarships, Department of Computer Science and Operations Research, Université de Montréal

2019–2020 Excellence Scholarship (Bourse C), Université de Montréal

2018 IAREP/SABE Fellowship, 2nd Summer School on Psychological Game Theory

2017, 2018 Scholarship, Key Program of the National Social Science Foundation of China (Prof. J. Cui)

2016 Master's Thesis Research Scholarship, University of Pisa

TALKS

Seminars and invited talks

2026 IESE Business School

2026 WU Vienna University of Economics and Business

2026 Universitat Pompeu Fabra, Barcelona

2026 CIRRELT

2026 FairPlay team (Inria, CREST, Criteo), Paris

Nov 2025 University of Michigan, Department of EECS (group of V. Subramanian), Ann Arbor, MI – *Sharing with Frictions*

Oct 2025 TOC for Fairness Seminar, Simons Collaboration on the Theory of Algorithmic Fairness (online) – *Sharing with Frictions*

May 2025 Theory Brown Bag Seminar, Northwestern University, Evanston, IL

Feb 2025 Dagstuhl Seminar 25071, *Dealing with Complexities in Auction and Matching Market Design*, Schloss Dagstuhl, Germany – *Sharing with Frictions*

Nov 2024 Quantitative Methods and Statistics Seminar, Purdue University, West Lafayette, IN

Feb 2024 MIT, Department of EECS (group of D. Shah), Cambridge, MA – *Dynamic Capacities and Priorities in Stable Matching* (Ph.D. thesis)

Oct 2023 CIREQ Microeconomics Seminar, Department of Economics, Université de Montréal

Nov 2022 International Symposium on Current Topics of Logical Structures in Mathematics, Ege University, Izmir, Turkey (invited speaker)

Conference and workshop presentations

Sep 2026 TPRC54, Research Conference on Communications, Information and Internet Policy, Washington, DC – *Market Signals without Mission Distortion*

Oct 2025 INFORMS Annual Meeting, Workshop on Replicability at INFORMS Journals, Atlanta, GA – *Capacity Planning in Stable Matching*

Oct 2025 INFORMS Annual Meeting, INFORMS Optimization Society (OPT) session on Nonlinear Optimization, Atlanta, GA

Oct 2025 SpectrumX Center Annual Meeting, University of Wisconsin–Madison, WI

Oct 2025	Young Researchers Workshop, Cornell University, Ithaca, NY – *Sharing with Frictions* (poster)
Aug 2025	Future BA Prof Workshop, University of Iowa, Iowa City, IA – *Sharing with Frictions*
Jul 2025	Workshop on Fairness in Social Choice, Northwestern University, Evanston, IL – *Stable Matching with Contingent Priorities*
Dec 2024	SpectrumX Policy Meeting (online)
Jul 2024	ISMP 2024, International Symposium on Mathematical Programming, Montréal – *Stable Matching with Contingent Priorities*
Nov 2023	SLMath workshop *Algorithms, Approximation, and Learning in Market and Mechanism Design*, Berkeley, CA – *Capacity Planning in Stable Matching* (poster)
Oct 2023	INFORMS Annual Meeting, Game Theory and Discrete Optimization session, Phoenix, AZ
Oct 2023	Young Researchers Workshop, Cornell University, Ithaca, NY – *Capacity Planning in Stable Matching* (poster)
Jul 2023	ACM EC'23, Conference on Economics and Computation, London – *Capacity Planning in Stable Matching*
Jun 2023	MSOM 2023, INFORMS Manufacturing and Service Operations Management Conference, Montréal – *Stable Matching with Contingent Priorities*
Oct 2022	ACM EAAMO'22, Equity and Access in Algorithms, Mechanisms, and Optimization, Arlington, VA – *Capacity Planning in Stable Matching*
Aug 2022	MATCH-UP 2022, Vienna – *Capacity Planning in Stable Matching*
May 2022	Mixed Integer Programming Workshop 2022, DIMACS, Rutgers University, NJ – *Margaridinhas vs. MIP 2022 Competition* (poster)
May 2022	GAIW 2022, Games, Agents, and Incentives Workshop – *Capacity Expansion in the College Admission Problem*
Nov 2021	Workshop on Agents' Behaviour in Combinatorial Game Theory, Montréal – *Capacity Expansion in the College Admission Problem* (poster)
Oct 2021	INFORMS Annual Meeting 2021 – *Capacity Expansion in the College Admission Problem*
Jul 2021	EURO 2021, 31st European Conference on Operational Research, Athens – *Capacity Expansion in the College Admission Problem*
Dec 2017	EUMAS 2017, European Conference on Multi-Agent Systems, Évry, France – *A Plausibility Model for Regret Games*
Aug 2017	Logic Colloquium 2017, Stockholm – *A Plausibility Model for Regret Games*
Jul 2017	Strategic Reasoning Workshop 2017, Liverpool – *A Plausibility Model for Regret Games*

TEACHING

Fall 2026	**Instructor**, SKEMA Business School, Lille Artificial Intelligence for Business and Society; Consumer Computer Applications; Decarbonization and Business (E3 initiative)
2021–2022	**Teaching Assistant**, Department of Computer Science and Operations Research, Université de Montréal Models of Operations Research (taught in French), Fall 2021, Winter 2022, Fall 2022; average student evaluation 3.9/4.0 in Fall 2022
2018	**Teaching Assistant**, Department of Decision Sciences, Bocconi University Game Theory, Spring 2018

RESEARCH SUPERVISION

2023	Marylou Fauchard, undergraduate student, Université de Montréal Research project: fairness in stable matching (weekly supervision)

2023 Alexandre Charland, undergraduate student, Université de Montréal
Research project: polynomial-time search of the lattice of stable matchings (biweekly supervision)

PROFESSIONAL SERVICE AND LEADERSHIP

2023 Treasurer, ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO'23)

2021–2024 Lead organizer, Coffee Talks on Machine Learning, Game Theory and Optimization, CERC Data Science for Real-Time Decision-Making, Montréal

Reviewing *Journals:* Theoretical Computer Science; Dynamic Games and Applications; Social Choice and Welfare
Conferences and workshops: ACM EC; IPCO; ACM EAAMO; ACM FAccT; MATCH-UP; GAIW

MEMBERSHIPS

2017 Member, Association for Symbolic Logic

OUTREACH AND MEDIA

Aug 2025 Article, *When Black Holes Meet Broadband: Incentivizing Honest Spectrum Sharing*, Daniels Insights, Daniels School of Business, Purdue University

Jun 2025 Featured in Northwestern ECE news: *Postdoc Federico Bobbio Wins Best Paper Award at IEEE DySPAN 2025*

SKILLS AND LANGUAGES

Computing Python (Gurobi, SCIP), Julia (Gurobi)

Languages Italian (native), English (proficient), French (advanced), Spanish (conversational)